1. What is Back of the Envelope (BOTE)?

Back of the Envelope estimation is a quick, rough calculation technique used in High-Level Design (HLD) interviews to assess system scale, resource requirements, and potential bottlenecks — *before* writing a single line of architecture.

① **Traffic & Throughput**
Calculate QPS (Queries Per Second) and network bandwidth. Reveals how many servers you need.

② **Storage**
Estimate raw data per day/month/year. Factors in text, media, and data growth.

③ **Memory & Cache**
Determine RAM required to cache hot data (typically 20% of requests drive 80% of load — Pareto principle).

2. Quick Reference Cheat Sheet

The core mental model: **Zeros** → **Traffic scale** → **Storage unit**. Memorise this table — it lets you convert any large number instantly.

Power	Traffic Scale	Storage Unit	Data Type	Size
10^3	1 Thousand	KB	char	2 bytes
10■	1 Million	MB	int / float	4 bytes
10■	1 Billion	GB	long / double	8 bytes
10^{12}	1 Trillion	TB	image (avg)	300 KB
10^{1}■	1 Quadrillion	PB	video (1 min)	~50 MB

Tip: 86,400 ≈ 10^5 seconds in a day — use this shortcut for QPS calculations.

3. Case Study — Facebook Daily Active Users

Assumptions	1. 1 Billion total users; **DAU = 250 Million** (25% active) 2. Each user performs **5 reads + 2 writes** per day (7 queries total) 3. Each user writes **2 posts × 250 chars** → 500 bytes/post 4. **10% of DAU** (25M users) post 1 image of 300 KB/day

3a. Estimating Traffic (QPS)

Calculation

```
Queries per user per day = 5 reads + 2 writes = 7 queries
Total queries per day = 250,000,000 × 7
= 1,750,000,000 (1.75 Billion queries/day)
Seconds in a day ≈ 86,400 (~10■)
```

=> QPS = 1,750,000,000 / 86,400

=> QPS ≈ 20,000 requests / second (~18K to 20K)

Peak QPS (2× average for traffic spikes) ≈ 36K–40K req/sec

Result: ~18K–20K QPS average | ~36K–40K peak QPS

3b. Estimating Storage

Text / Posts

```
1 char = 2 bytes → 250 chars = 500 bytes per post
2 posts per user → 1,000 bytes = 1 KB per user/day
```

=> 250M users × 1 KB = 250 GB / day (text storage)

Over 5 years ≈ 250 GB × 365 × 5 ≈ ~450 TB (text)

Daily text storage: 250 GB | 5-year total: ~450 TB

Images

```
10% of DAU post 1 image → 25,000,000 images/day
Avg image size = 300 KB
```

=> 25M × 300 KB = 7,500 GB ≈ 7.5 TB / day

Over 5 years ≈ 7.5 TB × 365 × 5 ≈ ~13.7 PB (images)

Daily image storage: ~7.5 TB | 5-year total: ~14 PB

Note: These figures exclude replication. In production, multiply by 3× (standard replication factor) for fault tolerance across data centres.

3c. Estimating RAM (Cache)

Calculation

```
Cache the last 5 posts per user in memory
1 post = 500 bytes → 5 posts = 2,500 bytes ≈ 3 KB per user
```

=> 250M users × 3 KB = 750,000,000 KB = 750 GB RAM needed

This is the 'hot data' cache — frequently read content

Total RAM required: ~750 GB

3d. Estimating Number of Servers

Memory Servers

```
Each server holds 75 GB in memory
```

=> 750 GB / 75 GB = 10 servers (for RAM/cache layer)

Application Servers (QPS-based)

```
Assumed server: 50 threads, each request latency = 500 ms
1 thread handles: 1000 ms / 500 ms = 2 requests/sec
1 server handles: 50 threads × 2 req/sec = 100 req/sec
Total QPS to serve: ~18,000 req/sec
```

=> Servers needed = 18,000 / 100 = 180 servers

Add 20% buffer for failures/upgrades → ~216 servers

Application servers: ~180 servers (216 with headroom)

4. Results Summary

Metric	Estimate
QPS (average / peak)	~18K / ~36K requests per second
Application servers	~180 servers (216 with 20% buffer)
RAM / cache servers	750 GB → 10 servers
Text storage (daily)	250 GB / day
Image storage (daily)	~7.5 TB / day
Total storage (5 years)	~450 TB (text) + ~14 PB (images)

5. Design Trade-off (CAP Theorem)

Facebook prioritises:	**Availability + Partition Tolerance** over strict Consistency. A like count being off by a few seconds is acceptable. A user *not* being able to post at all is not.

6. BOTE Interview Tips

State assumptions first	Always declare your assumptions out loud. The interviewer cares about your reasoning, not the exact number.
Use round numbers	3 Billion, not 2.98 Billion. Simplicity speeds up the math and reduces errors.
Factor in replication	Multiply raw storage by 3× for fault tolerance across data centres.
Peak vs average	Peak load is typically 2–3× average. Design for peak, budget for average.
Separate media from text	Images/videos dwarf text storage. Estimate them separately to avoid underestimating by orders of magnitude.
Know your powers of 10	Zeros → Traffic scale → Storage unit. This cheat sheet is your fastest tool in the interview.